Precomposition with an essentially surjective and full functor is fully faithful

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'ct:esofull-precomp-ff' (source lines 1390-1392, Lemma 9.9.2); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The displayed type carries the internal label eq:fullprop, which is defined and used inside the retained proof, so it needs no external entry.

Editorial note. The retained counter values (chapter 9, section 9, theorem offset 1) make the counted lemma print as Lemma 9.9.2, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial scope. One original result of the HoTT Book, counted once: precomposition with an essentially surjective and full functor is fully faithful (Lemma 9.9.2).

Allowed standard prerequisites (editorial).

- Fully faithful functor. A functor is fully faithful when its action on each hom-type is an equivalence.. It is the conclusion, and fullness is the part left to be shown here.
- Essentially surjective and full. H is essentially surjective and full when preimages exist up to truncation and the action on hom-types is surjective.. Both hypotheses are consumed in the proof, fullness at source lines 1405 and 1424.
- Contractibility is a mere property. To show a type is contractible it suffices to show it is inhabited and a mere proposition, and truncations may be stripped when the goal is a proposition.. It is the structure of the argument around the displayed type at source line 1397.

Lemma 9.9.2. *If A, B, C are precategories and $H : A \to B$ is essentially surjective and full, then $(- \circ H) : C^B \to C^A$ is fully faithful.*

Proof. It remains to show fullness. Thus, let $F, G : B \to C$ and $\gamma : FH \to GH$. We claim that for any $b : B$, the type

$$\sum_{(g:\hom_C(Fb,Gb))} \prod_{(a:A)} \prod_{(f:Ha\cong b)} (\gamma_a = Gf^{-1} \circ g \circ Ff) \tag{9.9.3}$$

is contractible. Since contractibility is a mere property, and H is essentially surjective, we may assume given $a_0 : A$ and $h : Ha_0 \cong b$.

Now take $g :\equiv Gh \circ \gamma_{a_0} \circ Fh^{-1}$. Then given any other $a : A$ and $f : Ha \cong b$, we must show $\gamma_a = Gf^{-1} \circ g \circ Ff$. Since H is full, there merely exists a morphism $k : \hom_A(a, a_0)$ such that $Hk = h^{-1} \circ f$. And since our goal is a mere proposition, we may assume given some such k. Then we have

$$\begin{aligned}\gamma_a &= GHk^{-1} \circ \gamma_{a_0} \circ FHk \\ &= Gf^{-1} \circ Gh \circ \gamma_{a_0} \circ Fh^{-1} \circ Ff \\ &= Gf^{-1} \circ g \circ Ff.\end{aligned}$$

Thus, (9.9.3) is inhabited. It remains to show it is a mere proposition. Let $g, g' : \hom_C(Fb, Gb)$ be such that for all $a : A$ and $f : Ha \cong b$, we have both $(\gamma_a = Gf^{-1} \circ g \circ Ff)$ and $(\gamma_a = Gf^{-1} \circ g' \circ Ff)$. The dependent product types are mere propositions, so all we have to prove is $g = g'$. But this is a mere proposition, so we may assume $a_0 : A$ and $h : Ha_0 \cong b$, in which case we have

$$g = Gh \circ \gamma_{a_0} \circ Fh^{-1} = g'.$$

This proves that (9.9.3) is contractible for all $b : B$. Now we define $\delta : F \to G$ by taking δ_b to be the unique g in (9.9.3) for that b. To see that this is natural, suppose given $f : \hom_B(b, b')$; we must show $Gf \circ \delta_b = \delta_{b'} \circ Ff$. As before, we may assume $a : A$ and $h : Ha \cong b$, and likewise $a' : A$ and $h' : Ha' \cong b'$. Since H is full as well as essentially surjective, we may also assume $k : \hom_A(a, a')$ with $Hk = h'^{-1} \circ f \circ h$.

Since γ is natural, $GHk \circ \gamma_a = \gamma_{a'} \circ FHk$. Using the definition of δ, we have

$$\begin{aligned}Gf \circ \delta_b &= Gf \circ Gh \circ \gamma_a \circ Fh^{-1} \\ &= Gh' \circ GHk \circ \gamma_a \circ Fh^{-1} \\ &= Gh' \circ \gamma_{a'} \circ FHk \circ Fh^{-1} \\ &= Gh' \circ \gamma_{a'} \circ Fh'^{-1} \circ Ff \\ &= \delta_{b'} \circ Ff.\end{aligned}$$

Thus, δ is natural. Finally, for any $a : A$, applying the definition of δ_{Ha} to a and 1_a, we obtain $\gamma_a = \delta_{Ha}$. Hence, $\delta \circ H = \gamma$. □